

Keeping up with changing standards

Designers, test labs and test equipment manufacturers should be very well aware of, and up-to-date, with the changing technology and standards.

Test equipment users must be familiar with the equipment provided by the vendors and their limitations. It is best to buy an instrument that offers programmability for test parameters. This helps you adjust the testing process to minor revisions of standards. Programmability also serves some customised, non-standard requirements that the customers would have adopted to make their products more reliable in a typical operating environment.

Test facility vendors may have to strengthen their equipment as per standards for the safety requirements. Most of the required upgrades would be with respect to sensitivity of the antennae, range of antennae, software updates of the test equipment and periodical calibrations.

"Specifically for the changing compliance standards, we proactively reach out to the designated authorities to know about the likely future compliance changes," says Raghu Rao, manager of application engineering team of Tektronix India. "This helps in being ahead in the market with future ready solutions."

Building a basic EMC troubleshooting kit

Troubleshooting is the process of isolating a problem and applying appropriate fixes. A test engineer diagnoses an EMI/EMC problem the same way a doctor diagnoses the medical condition of a patient. The diagnosis involves several stages: looking at clues, examining the equipment, gathering additional information (usually through tests) and finding a suitable solution.

An EMC troubleshooting kit is of great help to both independent and in-house EMC consultants. The essential tools for EMC troubleshooting depend on the kind of industry compliance the customer is looking for. But here is a brief guide that would help you

Whom to approach for testing in design stage

At the design stage itself, the user/designer has to estimate the size, power consumption, equipment required for testing and the range for which radiations and conductive emissions need to be tested. Vendors are selected depending on these parameters. For example, if a switch mode power supply (SMPS) or a radio system were to be tested, a small chamber with a minimum of Standardisation Testing and Quality Certification (STQC) Directorate approval would be sufficient. Volume of the chamber matters as the DUT should fit inside the closed chamber.

For preliminary tests, industries can develop their own test setups without any approval. This is done only for confirming the emission levels before taking an appointment with the certification body. Underwriters Laboratories (UL) facilitate testing and certification of standards for the industrial grade. But when it comes to testing of defence applications, one has to approach defence organisations like RCI in Hyderabad, who can test and certify for EMI/EMC and EMP (electromagnetic pulse).

Latest Standards for India

Product	Standard
Electronic games (video)	IS 616:2010
Laptops/Notebooks/Tablets	IS 13252:2010
Plasma /LCD /LED TVs of screen size 81cm and above	IS 616:2010
Optical disc players with built-in amplifiers (input power 200W and above)	IS 616:2010
Microwave ovens	IS 302-2:25:1994
Video monitors of screen size 81cm and above	IS 13252:2010
Printers, Plotters	IS 13252:2003
Scanners	IS 13252:2010
Wireless keyboards	IS 13252:2010
Telephone answering machines	IS 13252:2010
Amplifiers (input power 2000W and above)	IS 616:2010
Electronic musical systems (input power 200W and above)	IS 616:2010
Electronic clocks with mains power	IS 302-2:26:1994
Set-top boxes	IS 13252:2010
Automatic data processing machines	IS 13252:2010
<i>As per Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order 2012, issued by Department of Electronics and IT (DeitY), India</i>	

that are repetitive. The need for today's equipment is to be able to catch sporadic and rarely occurring events.

Probes. These include near-field probes and current probes. Probes can be constructed in lab with a little expertise in the field. H-field and E-field probes are near-field RF probes, constructed from semi-rigid coaxial cables, that are used to identify the source of electromagnetic interference emissions and potential radiation frequencies around circuits, cables and enclosures.

H-field probes are made basically from a

assemble the most basic equipments required for basic troubleshooting, depending on your requirements, at minimum possible cost. The kit can be used for limited pre-compliance testing and assessing radiated emissions.

Spectrum analyser. The heart of an EMC troubleshooting kit is a spectrum analyser. Look for an instrument that is not bulky, preferably a handheld analyser, as it would be easier to carry around. It should have good external command support, fast triggering speed and a capability to buffer many triggered measurements (gated sweep). More useful, if you can afford, is real-time spectrum analyser. Conventional EMC testing involves analysing events

conductive loop, and they detect magnetic fields produced by clock signals, control signals, serial data streams and switchers. A voltage is produced in the loop proportional to the magnetic field perpendicular to it. With larger size of loops you get higher sensitivity, but lower resolution. E-field probes, which make direct contact with the circuit, are used to find emissions on individual pins or PCB traces.

Poorly-bonded cables are said to be the major cause of radiated emission failures. A current probe, one of the most used accessories in troubleshooting, lets you measure the common-mode current flowing in wires or cables and identify a bad termination, which causes current leakage. Probes let you